

# Nikita Kazeev

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*Research Scientist at the intersection of AI and Physics*

## WORK EXPERIENCE

**CONSTRUCTOR GROUP** *Machine Learning Consultant* 2021–present

**NATIONAL UNIVERSITY OF SINGAPORE** *Postdoc under [Kostya Novoselov](#)* 2022–present

**Wyckoff Transformer for generating symmetric crystals** [ICML 2025](#) · [GitHub](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference by 4 orders of magnitude, to 0.05 GPU ms per structure
- Implemented the model in PyTorch; production-ready package published on [PyPI](#)
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

**ML for defects in 2D crystals** [npj 2D Mater. Appl.](#) · [npj Comput. Mater.](#) · [patent](#) · [GitHub](#)

- Conceived a novel sparse representation of crystals with defects
- Achieved a 3.7x energy prediction error reduction
- Optimized training and inference: 4x less memory, 8x fewer GPU operations
- Developed code for reproducible parallel running of machine learning experiments
- Led a team of 3

**CERN [YANDEX → HSE UNIVERSITY]** *Researcher* 2014–2022

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

**Generative models for fast simulation** [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately  $10^5$  speed-up compared to the ab-initio simulator.

**Machine learning on noisy data** [paper 1](#) · [paper 2](#) · [code](#)

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

**Muon identification at the LHCb experiment at CERN** [paper](#) · [poster](#) · [IDAO-2019](#)

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

## EDUCATION

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**Higher School of Economics (HSE University)** 2016–2020  
*PhD in Computer Science*, supervisor [Andrey Ustyuzhanin](#); [Machine Learning for particle identification in the LHCb detector](#).

**Sapienza – Università di Roma** 2016–2020  
*PhD in Physics* (double degree with HSE), supervisor [Barbara Sciascia](#).

**Product Management course at Yandex** 2018  
Focused coursework in product strategy and execution.

**Yandex School of Data Analysis** 2013–2015  
Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

**Moscow Institute of Physics and Technology**  
*MS in Physics* (2014–2016): Optimisation of data processing of the LHCb experiment.  
*BS in Physics* (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

**International Junior Science Olympiad (IJSO)** 2008  
Silver medal (held in Korea).

## MENTORSHIP

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- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

## TEACHING & OUTREACH

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- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
- Delivered more than 20 [conference talks](#).

## SERVICE

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- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

## SKILLS

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- **ML & AI** – Structured, non-text, domain-specific data
- **Python** – Research coding
- **PyTorch** – Deep learning
- **C++** – Production
- **Linux** – Administration
- **Research Writing & Speaking**
- **Physics** – Deep domain understanding
- **Leadership & Mentorship**

## SELECTED PUBLICATIONS

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1. **Kazeev, Nikita**, and Ian Babich. [Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#). Forecasting as a New Frontier of Intelligence Workshop at ICML 2026
2. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Bundenyy, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#). AI for Science Workshop at ICML 2026
3. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. [Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement](#). arXiv preprint arXiv:2605.14759, May 2026.
4. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourrier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
5. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
6. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). npj Comput Mater 9, 113 (2023).
7. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. Alphabetic order, I'm the corresponding author.